Tangents and normals

1 $m = -\frac{1}{2}$

2

a $y = 5\pi x - 5\pi^2$

b $y = -3\sqrt{3}x + \frac{\sqrt{3}\pi}{6} + \frac{1}{2}$

3

a $x = \frac{\pi}{12}$

b $f'\left(\frac{\pi}{12}\right) = -4$

c $\theta = 104^\circ$

4 $\theta = 83^\circ$

5

a

i $\frac{dy}{dx} = 2\cos 2x$

ii Undefined

iii $x = \frac{\pi}{4}$

b

i $\frac{dy}{dx} = \sqrt{2}\cos x$

ii -1

iii $y = -x + \frac{\pi}{4} + 1$

c

i $\frac{dy}{dx} = -\sqrt{2}\sin x$

ii 1

iii $y = x - \frac{\pi}{4} + 1$

6

a $\frac{dy}{dx} = \frac{x\cos x - \sin x}{x^2}$

b $\frac{\pi^2}{4}$

c $y = \frac{\pi^2 x}{4} - \frac{\pi^3}{8} + \frac{2}{\pi}$

Trigonometric graphs

7

a $\frac{dy}{dx} = \frac{1}{1 + \cos x}$

b 0

8 $x = \frac{\pi}{8}, \frac{5\pi}{8}, \frac{9\pi}{8}, \frac{13\pi}{8}$

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9

a $\left(\frac{\pi}{3}, -4\right), \left(\frac{4\pi}{3}, 4\right)$

b $f''(x) = 4\sin\left(x + \frac{\pi}{6}\right)$

c $\left(\frac{\pi}{3}, -4\right)$ is a minimum, and $\left(\frac{4\pi}{3}, 4\right)$ is a maximum

10

a

i $y = 0, x = 0, \pi, 2\pi$

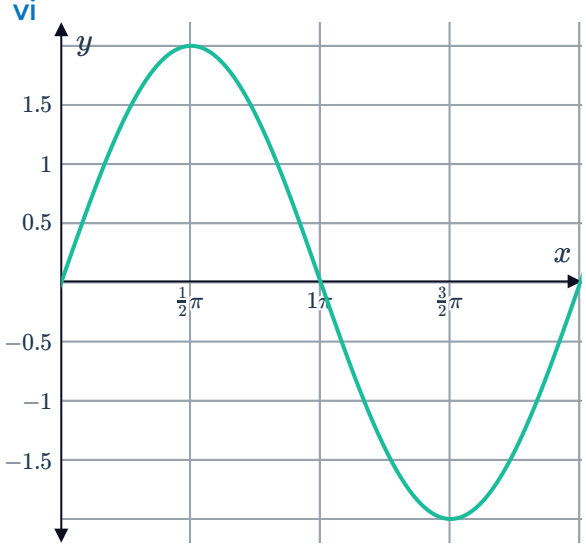
ii $f'(x) = 2\cos x$

iii $\left(\frac{\pi}{2}, 2\right), \left(\frac{3\pi}{2}, -2\right)$

iv $f''(x) = -2\sin x$

v Min at $\left(\frac{3\pi}{2}, -2\right)$, Max at $\left(\frac{\pi}{2}, 2\right)$

vi



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b

i $y = 1, x = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{3\pi}{2}, \frac{11\pi}{6}$

ii $f'(x) = -3\sin 3x$

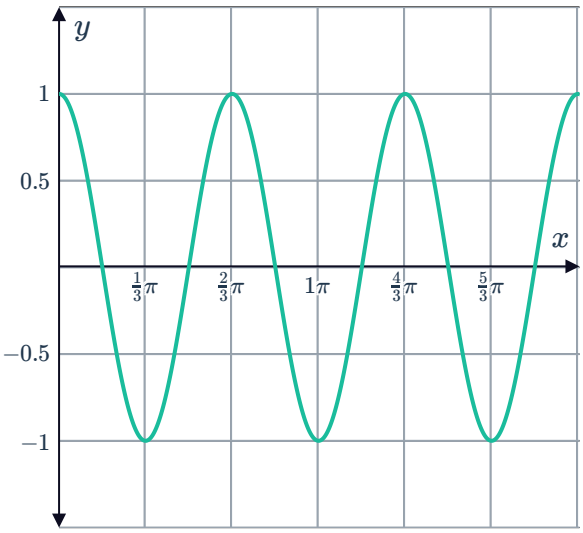
iii $\left(\frac{\pi}{3}, -1\right), (\pi, -1), \left(\frac{5\pi}{3}, -1\right), (0, 1), \left(\frac{2\pi}{3}, 1\right), \left(\frac{4\pi}{3}, 1\right), (2\pi, 1)$

iv $f''(x) = -9\cos 3x$

v Minimum at $\left(\frac{\pi}{3}, -1\right), (\pi, -1), \left(\frac{5\pi}{3}, -1\right)$

Maximum at $(0, 1), \left(\frac{2\pi}{3}, 1\right), \left(\frac{4\pi}{3}, 1\right), (2\pi, 1)$

vi



c

i $y = 0, x = 0, \frac{\pi}{2}, \frac{3\pi}{2}$

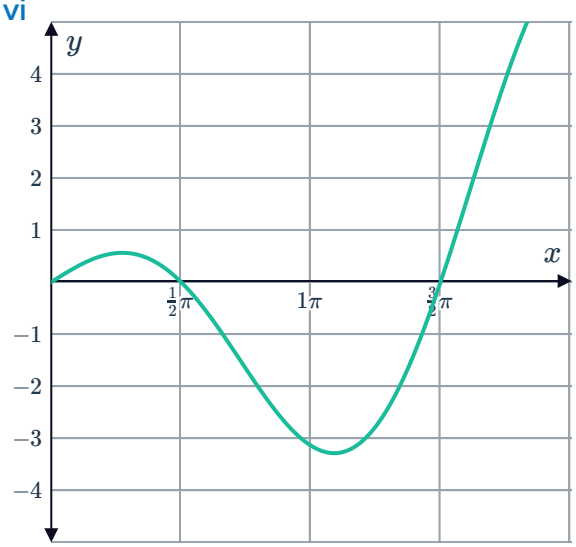
ii $f'(x) = \cos x - x\sin x$

iii $(0.86, 0.56), (3.42, -3.29)$

iv $f''(x) = -2\sin x - x\cos x$

v Minimum at $(3.42, -3.29)$
Maximum at $(0.86, 0.56)$

vi



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d

i $y = 1, x = \frac{\pi}{2}, \frac{3\pi}{2}$

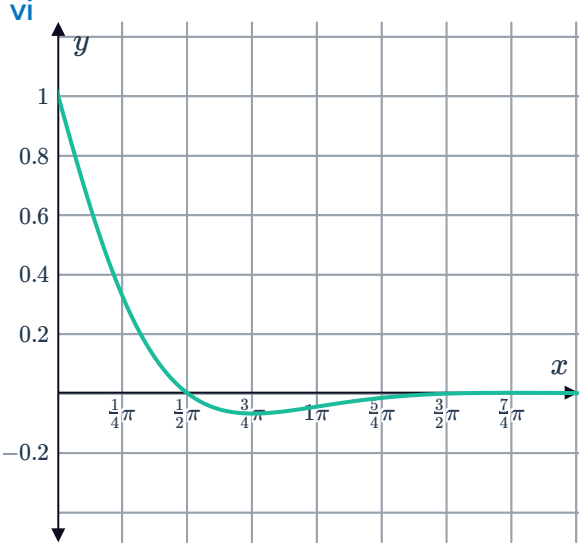
ii $f'(x) = -e^{-x}\sin x - e^{-x}\cos x$

iii $\left(\frac{3\pi}{4}, -\frac{e^{-\frac{3\pi}{4}}}{\sqrt{2}}\right)$

iv $f''(x) = 2e^{-x}\sin x$

v Minimum point

vi



e

i $y = 0, x = 0, \pi, 2\pi$

ii $f'(x) = 2\sin x\cos x$

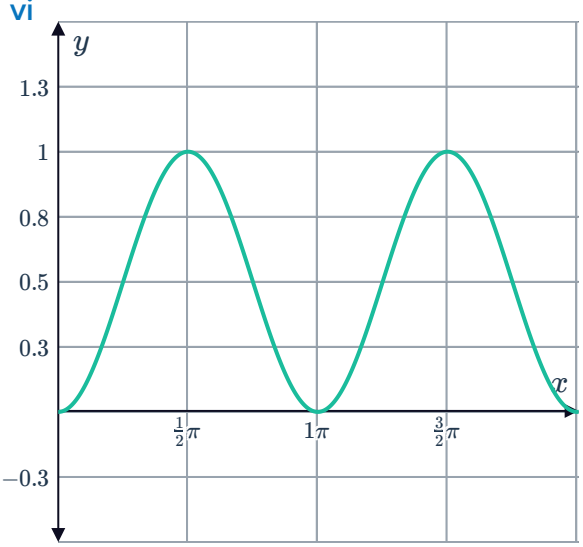
iii $(0,0), (\pi,0), (2\pi,0), \left(\frac{\pi}{2},1\right), \left(\frac{3\pi}{2},1\right)$

iv $f''(x) = 2\left(-\sin^2(x) + \cos^2(x)\right)$

v Minimum points: $(0,0), (\pi,0), (2\pi,0)$

Maximum points: $\left(\frac{\pi}{2},1\right), \left(\frac{3\pi}{2},1\right)$

vi



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f

i $y = 1, x = \frac{\pi}{2}, \frac{3\pi}{2}$

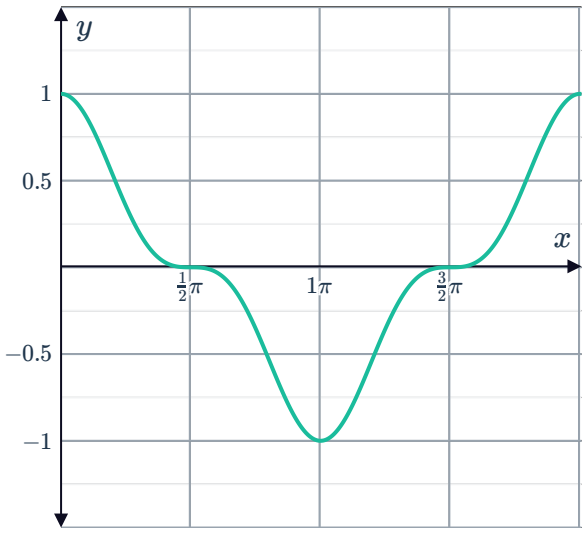
ii $f'(x) = -3\cos^2(x)\sin x$

iii $(\pi, -1), (0, 1), (2\pi, 1), \left(\frac{\pi}{2}, 0\right), \left(\frac{3\pi}{2}, 0\right)$

iv $f''(x) = -3\cos x\left(\cos^2(x) - 2\sin^2(x)\right)$

v Minimum point: $(\pi, -1)$, Maximum points: $(0, 1), (2\pi, 1)$
Horizontal points of inflection: $\left(\frac{\pi}{2}, 0\right), \left(\frac{3\pi}{2}, 0\right)$

vi



g

i $y = 0, x = 0$

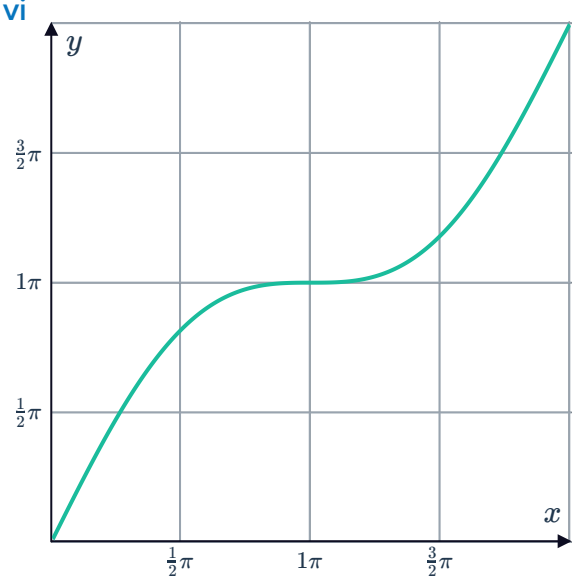
ii $f'(x) = 1 + \cos x$

iii (π, π)

iv $f''(x) = -\sin x$

v Horizontal point of inflection

vi



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Applications

11 $k = \pm 5$

12

a It is 5 units to the left of the origin.

b $v\left(t\right) = 10\sin 2t$

c No solution provided.

d $-20\ \text{cm/s}^2$

13

a $0\ \text{cm}$

b $v\left(t\right) = -2\cos^2\left(t\right) + \cos t - 4$

c $-5\ \text{cm/s}$

14

a $0\ \text{cm}$

b $16\ \text{cm}$

c $v\left(t\right) = 48\pi\cos 3\pi t$

d $t = \dfrac{1}{6}, \dfrac{1}{2}$

e $16\ \text{cm}$

f $-16\ \text{cm}$

g $32\ \text{cm}$

15

a $v\left(t\right) = 76\pi\cos 4\pi t$

b $76\pi\ \text{cm/s}$

c $t = \dfrac{1}{4}, \dfrac{1}{2}$

d The centre of its swing at $x = 0$.

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